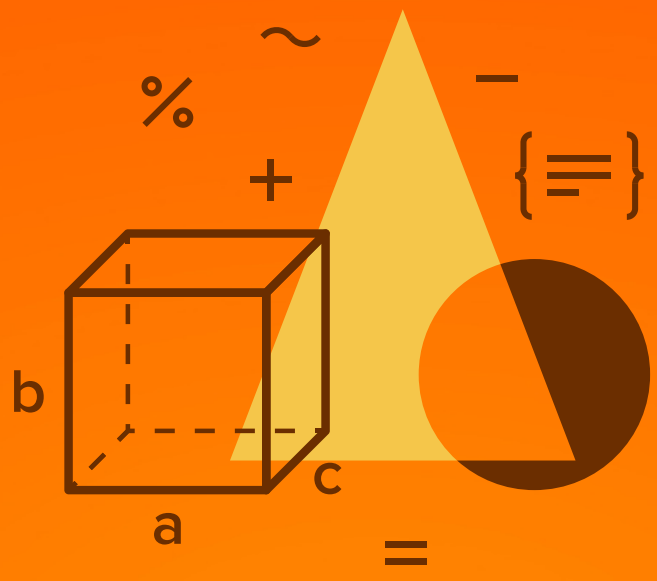
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MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

JEE Main Topics (Statistics & Reasoning)

English Medium

EXERCISE (O-1)

Straight Objective Type

1. If R is a relation from a finite set A having m elements to a finite set B having n elements, then the number of relations from A to B is-

(A) 2^{mn} (B) $2^{mn}-1$ (C) 2mn (D) m^n

RT0001

2. In the set A = {1, 2, 3, 4, 5}, a relation R is defined by R = {(x, y) | x, y ∈ A and x < y}. Then R is-

(A) Reflexive (B) Symmetric (C) Transitive (D) None of these

RT0002

3. For real numbers x and y, we write x R y ⇔ $x - y + \sqrt{2}$ is an irrational number. Then the relation R is-

(A) Reflexive (B) Symmetric (C) Transitive (D) none of these

RT0003

4. Let X = {1, 2, 3, 4} and Y = {1, 3, 5, 7, 9}. Which of the following is relations from X to Y-

(A) R_1 = {(x, y) | y = 2 + x, x ∈ X, y ∈ Y} (B) R_2 = {(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)}

(C) R_3 = {(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)} (D) R_4 = {(1, 3), (2, 5), (2, 4), (7, 9)}

RT0004

5. If R be a relation '<' from A = {1, 2, 3, 4} to B = {1, 3, 5} i.e. (a, b) ∈ R iff a < b, then RoR^{-1} is-

(A) {(1, 3), (1, 5), (2, 3), (2, 5), (3, 5), (4, 5)} (B) {(3, 1), (5, 1), (3, 2), (5, 2), (5, 3), (5, 4)}

(C) {(3, 3), (3, 5), (5, 3), (5, 5)} (D) {(3, 3), (3, 4), (4, 5)} **RT0062**

6. Let R be a relation defined in the set of real numbers by a R b ⇔ 1 + ab > 0. Then R is-

(A) Equivalence relation (B) Transitive (C) Symmetric (D) Anti-symmetric

RT0006

7. Which one of the following relations on R is equivalence relation-

(A) x R_1 y ⇔ |x| = |y| (B) x R_2 y ⇔ x ≥ y (C) x R_3 y ⇔ x | y (D) x R_4 y ⇔ x < y

RT0007

8. Two points P and Q in a plane are related if OP = OQ, where O is a fixed point. This relation is-

(A) Reflexive but not symmetric (B) Symmetric but not transitive

(C) An equivalence relation (D) none of these

RT0008

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9. Let R be a relation on the set of natural numbers N be xRy iff $2x + 3y = 21$, then R is

(A) reflexive (B) symmetric (C) transitive (D) none of these

10. Let a relation R is the set N of natural numbers be defined as $(x, y) \in R$ if and only if $x^2 - 4xy + 3y^2 = 0$ for all $x, y \in N$. The relation R is-

(A) Reflexive (B) Symmetric

(C) Transitive (D) An equivalence relation

RT0010

11. Let R be a relation defined on a set of real numbers such that a R b and if $|a| \leq b$, then relation R is :-

(A) Reflexive (B) Symmetric (C) Transitive (D) Equivalence

12. If $A = \{2, 3\}$ and $B = \{1, 2\}$, then $A \times B$ is equal to-

(A) $\{(2, 1), (2, 2), (3, 1), (3, 2)\}$ (B) $\{(1, 2), (1, 3), (2, 2), (2, 3)\}$

(C) $\{(2, 1), (3, 2)\}$ (D) $\{(1, 2), (2, 3)\}$

RT0012

13. Let R be a relation over the set $N \times N$ and it is defined by $(a, b) R (c, d) \Rightarrow a + d = b + c$. Then R is-

(A) Reflexive only (B) Symmetric only

(C) Transitive only (D) An equivalence relation

RT0013

14. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by (a, b) R (c, d) if $ad(b + c) = bc(a + d)$, then R is-

(A) Symmetric only (B) Reflexive only

(C) Transitive only (D) An equivalence relation

RT0014

15. If $A = \{1, 2, 3\}$, $B = \{1, 4, 6, 9\}$ and R is a relation from A to B defined by 'x is greater than y'. Then range of R is-

(A) $\{1, 4, 6, 9\}$ (B) $\{4, 6, 9\}$ (C) $\{1\}$ (D) none of these

RT0015

16. Let R and S be two equivalence relations in a set A. Then-

(A) $R \cup S$ is an equivalence relation in A

(B) $R \cap S$ is an equivalence relation in A

(C) $R - S$ is an equivalence relation in A

(D) none of these **RT0063**

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17. The relation R is defined on the set of natural numbers as $\{(a,b) : a = 2b\}$. Then R^{-1} is given by

(A) {(2, 1), (4, 2), (6, 3).....}
(B) {(1, 2), (2, 4), (3, 6)....}
(C) R^{-1} is not defined
(D) None of these

RT0064

18. For n, m ∈ N, n|m means that n is a factor of m, the relation | is-

(A) reflexive and symmetric
(B) transitive and symmetric
(C) reflexive, transitive and symmetric
(D) reflexive, transitive and not symmetric

RT0018

19. Let $R = \{(x, y) : x, y \in A, x + y = 5\}$ where $A = \{1, 2, 3, 4, 5\}$ then

(A) R is not reflexive, symmetric and not transitive
(B) R is an equivalence relation
(C) R is reflexive, symmetric but not transitive
(D) R is not reflexive, not symmetric but transitive

RT0019

20. Let R be a relation on a set A such that $R = R^{-1}$ then R is -

(A) reflexive (B) symmetric (C) transitive (D) none of these

RT0020

21. Let x, y ∈ I and suppose that a relation R on I is defined by x R y if and only if $x \le y$ then

(A) R is partial order relation
(B) R is an equivalence relation
(C) R is reflexive and symmetric
(D) R is symmetric and transitive

RT0021

22. Let R be a relation from a set A to a set B, then-

(A) $R = A \cup B$ (B) $R = A \cap B$ (C) $R \subseteq A \times B$ (D) $R \subseteq B \times A$

RT0022

23. Given the relation $R = \{(1, 2), (2, 3)\}$ on the set $A = \{1, 2, 3\}$, the minimum number of ordered pairs which when added to R make it an equivalence relation is-

(A) 5 (B) 6 (C) 7 (D) 8

RT0023

24. Let $P = \{(x, y) \mid x^2 + y^2 = 1, x, y \in R\}$ Then P is-

(A) reflexive (B) symmetric (C) transitive (D) anti-symmetric

RT0024

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25. Let X be a family of sets and R be a relation on X defined by 'A is disjoint from B'. Then R is-

(A) reflexive (B) symmetric (C) anti-symmetric (D) transitive

RT0025

26. In order that a relation R defined in a non-empty set A is an equivalence relation, it is sufficient thatR

(A) is reflexive (B) is symmetric

(C) is transitive (D) possesses all the above three properties

RT0026

27. If R is an equivalence relation in a set A, then R^{-1} is-

(A) reflexive but not symmetric (B) symmetric but not transitive

(C) an equivalence relation (D) none of these

RT0027

28. Let A = {p, q, r}. Which of the following is an equivalence relation in A ?

(A) $R_1 = \{(p, q), (q, r), (p, r), (p, p)\}$ (B) $R_2 = \{(r, q) (r, p), (r, r), (q, q)\}$

(C) $R_3 = \{(p, p), (q, q), (r, r), (p, q)\}$ (D) none of these

RT0028

29. The relation R = {(1, 1), (1, 2), (2, 1)} on the set {1, 2, 3} is

(A) Reflexive (B) Transitive (C) symmetric (D) none of these

RT0065

30. Let M denotes set of all 3×3 non singular matrices. Define the relation R by

$R = \{(A,B) \in M \times M : AB = BA\}$, then R is-

(A) Reflexive, symmetric but not transitive

(B) Reflexive, symmetric & transitive

(C) Reflexive, transitive but not symmetric **RT0066**

(D) Neither reflexive nor symmetric nor transitive

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EXERCISE (O-2)

Multiple Correct Answer Type

1. Let r be a relation from R (Set of real number) to R defined by r = {(x, y) | x, y ∈ R and xy is an irrational number}, then which of the following is/are incorrect -

(A) relation is reflexive and symmetric only (B) relation is symmetric only

(C) relation is symmetric and transitive only (D) relation is an equivalence relation **RT0067**

2. Let S = {(x, y)|siny = sinx, x, y ∈ R}, then S is

(A) transitive (B) not transitive

(C) reflexive (D) partial order relation **RT0068**

3. Let S be the set of all subsets of first 10 natural numbers. If a relation R is defined on set S such that (A,B) ∈ R if $A \cap B \neq \phi$; A, B ∈ S, then which of the following is/are incorrect -

(A) relation is symmetric and transitive (B) relation is symmetric and reflexive only

(C) relation is symmetric only (D) relation is reflexive only **RT0069**

4. If R and S are two non void relations on a set A. Then, which of the following statements is/are correct ?

(A) R and S are transitive implies $R \cap S$ is transistive

(B) R and S are transitive implies $R \cup S$ is transistive

(C) R and S are symmetric implies $R \cup S$ is symmetric

(D) R and S are reflexive implies $R \cap S$ is reflexive **RT0070**

5. Let N be the set of natural numbers greater than 100. Define the relation R by :

R = {(x,y) ∈ N × N : the numbers x and y have atleast two common divisors}. Then R is-

(A) Reflexive relation (B) Symmetric relation

(C) transitive relation (D) Equivalence relation **RT0071**

6. The relation R defined in A = {1, 2, 3} by a R b if $|a^2 - b^2| \leq 5$. Which of the following is/are true -

(A) R = {(1, 1), (2, 2), (3, 3), (2, 1), (1, 2), (2, 3), (3,2)

(B) $R^{-1} = R$

(C) Domain of R = {1, 2, 3}

(D) Range of R = {5} **RT0009**

7. Let A = {2, 3, 4, 5} and let R = {(2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (3, 5), (5, 3)} be a relation in A. Then R is-

(A) Reflexive relation (B) symmetric relation

(C) antisymmetric relation (D) none of these

RT0011

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8. A relation R on the set of complex numbers is defined by z_1 R z_2 if and only if $\frac{z_1 - z_2}{z_1 + z_2}$ is real, then R is

(A) Reflexive relation (B) Symmetric relation

(C) transitive relation (D) not transitive relation **RT0072**

9. Let S be a set of all square matrices of order 2. If a relation R defined on set S such that AR B $\Rightarrow$ AB = O, where O is zero square matrix of order 2, then relation R is (A, B $\in$ S)

(A) Not reflexive (B) Transitive

(C) Symmetric (D) Not equivalence **RT0073**

10. Let A = N × N be the Cartesian product of N and N. Let

S = {((m, n), (p, q)) $\in$ A × A : m + q = n + p}

Consider the following statements:

I. If ((m,n), (p , q)) $\in$ S, and ((p,q), (r, s)) $\in$ S then ((r,s), (m,n)) $\in$ S

II. There exists at least one element ((m,n), (p, q)) $\in$ S such that ((p , q), (m, n)) $\notin$ S

Then select false options ?

(A) I is correct (B) II is correct

(C) Both I and II are correct (D) Neither I nor II is correct **RT0074**

11. Let A = Z, the set of integers. Let R_1 = {(m, n) $\in$ Z × Z : (m + 4n) is divisible by 5 in Z}.

Let R_2 = {(m, n) $\in$ Z × Z : (m + 9n) is divisible by 5 in Z}.

Which of the following is/are incorrect ?

(A) R_1 is a proper subset of R_2

(B) R_2 is a proper subset of R_1

(C) $R_1 = R_2$

(D) R_1 is not a symmetric relation on Z **RT0075**

12. Let X be the set of all persons living in a state. Elements x, y in X are said to be related if 'x < y', whenever y is 5 years older than x. Which of the following is/are incorrect?

(A) The relation is an equivalence relation

(B) The relation is transitive only

(C) The relation is transitive and symmetric, but not reflexive

(D) The relation is neither reflexive, nor symmetric, nor transitive **RT0076**

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13. Let N be the set of natural numbers and a relation R on N be defined by

$R = \{(x,y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is :

(A) symmetric but neither reflexive nor transitive

(B) reflexive

(C) not transitive

(D) an equivalence relation **RT0077**

14. Which of the following is/are correct for relation R on the set of real numbers ?

(A) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

(B) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.

(C) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.

(D) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric. **RT0078**

15. Define a relation R over a class of $n \times n$ real matrices A and B as "ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$". Then which of the following is/are false ?

(A) R is symmetric, transitive but not reflexive,

(B) R is reflexive, symmetric but not transitive

(C) R is an equivalence relation

(D) R is reflexive, transitive but not symmetric **RT0079**

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for question nos. 1 to 3

A relation R is defined on set A = {1,2,3...,n}.

1. Total number of reflexive relations that can be defined from A → A is- **RT0080**

(A) 2^{n^2-n} (B) 2^{n^2} (C) 2^{n^2-n+1} (D) 2^{n^2+n}

2. Find the maximum number of antisymmetric relation in R is **RT0081**

(A) $3^{\frac{n^2-n}{2}}.2^n$ (B) $3^{\frac{n^2-n}{2}}$ (C) $3^{\frac{n^2+n}{2}}.2^n$ (D) 2^{n^2}

3. Total number of symmetric relations that can be defined from A → A is- **RT0082**

(A) $2^{\frac{n(n+2)}{2}}$ (B) $2^{\frac{n(n-1)}{2}}$ (C) $2^{\frac{n(n+1)}{2}}$ (D) $2^{n(n+2)}$

Paragraph for question nos. 4 and 5

Given A = {1, 2, 3} ; B = {3, 4, 5} ; C = {2, 4, 6} then

4. Total number of relation possible from (A – B) to (B – C) is greater than **RT0083**

(A) 7 (B) 3 (C) 15 (D) 23

5. Total number of relation possible from $((A \cup B) \cap C)$ to $(A \times B)$ is 2^m then m is less than

(A) 33 (B) 9 (C) 16 (D) 19 **RT0084**

Paragraph for question nos. 6 to 8

Let A = {1,2,3} and relation R be defined on A as follows :

R = {(1,1), (1,2), (1,3)}

6. Find the minimum number of ordered pairs to be added in R to make it reflexive relation.

RT0085

7. Find the minimum number of ordered pairs to be added in R to make it symmetric relation

RT0086

8. Find the minimum number of ordered pairs to be added in R to make it equivalence relation.

RT0087

9. Match the relation defined on set A = {a,b,c} in List-I with the corresponding type in List-II

List-I		List-II	
(I)	{(a,b), (b,a)}	(P)	symmetric but not reflexive and transitive
(II)	{(a,b), (b,a), (a,a), (b,b)}	(Q)	equivalence
(III)	{(a,b), (b,c), (a,c)}	(R)	symmetric and transitive but not reflexive
(IV)	{(a,a), (b,b), (c,c)}	(S)	transitive but not reflexive and symmetric

(A) I → P; II → Q; III → R; IV → S (B) I → P; II → R; III → S; IV → Q

(C) I → R; II → S; III → P; IV → Q (D) I → Q; II → R; III → S; IV → P

RT0088

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10.

	Column-I		**Column-II**
(A)	$R = \{(x,y) \in \mathbb{R} \times \mathbb{R}, (x - y) \text{ is an integer}\}$, is-	(P)	Reflexive relation
(B)	$R = \{(x,y) \in \mathbb{R} \times \mathbb{R}, \lvert x - y\rvert \le \frac{1}{2}\}$, is-	(Q)	Symmetric relation
(C)	Let I be the set of positive integers. $R = \{(a,b) \in I \times I \mid \log_2\left(\frac{a}{b}\right) \text{ is a non-negative integer}\}$, is	(R)	Transitive relation
(D)	$R = \{(6, 6), (9, 9), (6, 12), (12, 12), (12,6)\}$ is a relation on set $A = \{3, 6, 9, 12\}$, is -	(S)	Equivalence relation

RT0089

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EXERCISE (O-4)

Numerical Grid Type

1. If $A = \{x : x^2 - 5x + 6 = 0\}$, $B = \{2, 4\}$, $C = \{4, 5\}$ then find number of relation from $A \times (B \cap C)$ to $(B \cup C)$. **RT0090**

2. The Cartesian product $A \times A$ has 16 elements. $R = \{(a,b) \in A \times A \mid a < b\}$. $(-1,2)$ and $(0,1)$ are two elements belonging to R. Number of remaining elements of Relation R is **RT0091**

3. If $A = \{2, 4, 6, 9\}$ and $B = \{4, 6, 18, 27, 54\}$, $a \in A$, $b \in B$. Find number of elements in Relation R such that $R = \{(a, b) \in A \times B : a \text{ is a factor of } b, a < b\}$. **RT0092**

4. If $P = \{x : x < 3, x \in \mathbb{N}\}$, $Q = \{x : x \le 2, x \in W\}$. Find total number of subset of $(P \cup Q) \times (P \cap Q)$, where W is the set of whole numbers. **RT0093**

5. If R is a relation on the set $\mathbb{N}$ (where $\mathbb{N}$ is set of natural number), defined by $\{(x,y) \in \mathbb{N} \times \mathbb{N} : 3x + 3y = 10\}$-

Statement -1 : R is symmetric

Statement -2 : R is reflexive

Statement -3 : R is transitive, then number of correct statements. **RT0094**

6. Let $A = \{a,b,c,d\}$ and R be a reflexive relation.

α = minimum number of elements in R

β = maximum number of elements in R

Then $(\alpha + \beta)$ is : **RT0095**

7. Let R be a relation defined as $R = \{(x, y) : y = \sqrt{(x-1)^2}, x \in Z \text{ and } -3 \le x \le 3\}$ then number of elements in range set. **RT0096**

8. A and B are two sets having 3 and 4 elements respectively and having 2 elements in common. Find the number of possible relations which can be defined from A to B. **RT0097**

9. Let $A = \{p, q, r\}$. How many equivalence relation on A ?

(i) $R = \{(p, q), (q, r), (p, r) (p, p)\}$

(ii) $R = \{(p, p), (q, q), (r, r) (q, p)\}$

(iii) $R = \{(p, p), (q, q), (r, r)\}$

(iv) $R = \{(p, p), (q, q), (r, r), (p, q), (q, r), (p, r)\}$

(v) $R = \{(p, p), (q, q), (r, r), (p, q), (q, p)\}$ **RT0098**

10. Consider the following :

1. If $R = \{(a, b) \in N \times N : a \text{ divides } b \text{ in } N\}$ then the relation R is reflexive and symmetric but not transitive.
2. If $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(S_1, S_2) : S_1, S_2 \text{ are subsets of } A, S_1 \not\subset S_2\}$, then the relation R is not reflexive, not symmetric and not transitive.

Number of correct statements. **RT0099**

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EXERCISE (JM)

1. Let R = {(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)} be a relation on the set A = {1, 2, 3, 4}. The relation R is-

[AIEEE - 2004]

(1) transitive (2) not symmetric (3) reflexive (4) a function

RT0029

2. Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be relation on the set A = {3, 6, 9, 12). The relation is - **[AIEEE - 2005]**

(1) reflexive and transitive only (2) reflexive only

(3) an equilvalence relation (4) reflexive and symmetric only

RT0030

3. Let W denote the words in the English dictionary. Define the relation R by : R = {(x, y) ∈ W × W| the words x and y have at least one letter in common}. Then R is- **[AIEEE - 2006]**

(1) reflexive, symmetric and not transitive

(2) reflexive, symmetric and transitive

(3) reflexive, not symmetric and transtive

(4) not reflexive, symmetric and transitive

RT0031

4. Consider the following relations :-

R = {(x, y) | x, y are real numbers and x = wy for some rational number w} ;

$S = \{(\frac{m}{n}, \frac{p}{q}) \mid m, n, p \text{ and } q \text{ are integers such that } n, q \neq 0 \text{ and } qm = pn\}$.

Then : **[AIEEE - 2010]**

(1) R is an equivalence relation but S is not an equivalence relation

(2) Neither R nor S is an equivalence relation

(3) S is an equivalence relation but R is not an equivalence relation

(4) R and S both are equivalence relations

RT0032

5. Let R be the set of real numbers. **[AIEEE - 2011]**

Statement-1:

A = {(x, y) ∈ R × R : y – x is an integer} is an equivalence relation on R.

Statement-2:

B = {(x, y)∈ R × R : x = αy for some rational number α} is an equivalence relation on R.

(1) Statement-1 is true, Statement-2 is false.

(2) Statement-1 is false, Statement-2 is true.

(3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(4) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.

RT0033

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6. If $R = \{(x, y) : x, y \in Z, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers Z, then the domain of R^{-1} is : **[JEE(Main) 2020]**

(1) $\{-2, -1, 1, 2\}$ (2) $\{-1, 0, 1\}$

(3) $\{-2, -1, 0, 1, 2\}$ (4) $\{0, 1\}$ **RT0100**

7. Let R_1 and R_2 be two relations defined as follows :

$R_1 = \{(a, b) \in R^2 : a^2 + b^2 \in Q\}$ and

$R_2 = \{(a, b) \in R^2 : a^2 + b^2 \notin Q\}$,

where Q is the set of all rational numbers. Then : **[JEE(Main) 2020]**

(1) R_2 is transitive but R_1 is not transitive

(2) R_1 is transitive but R_2 is not transitive

(3) R_1 and R_2 are both transitive

(4) Neither R_1 nor R_2 is transitive **RT0101**

8. Let $A = \{2, 3, 4, 5, ... , 30\}$ and '$\simeq$' be an equivalence relation on $A \times A$, defined by $(a,b) \simeq (c,d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to : **[JEE(Main) 2021]**

(1) 5 (2) 6 (3) 8 (4) 7 **RT0102**

9. Let R_1 and R_2 be two relations defined on $\mathbb{R}$ by

$a\ R_1\ b \Leftrightarrow ab \geq 0$ and $a\ R_2\ b \Leftrightarrow a \geq b$, then **[JEE(Main) 2022]**

(1) R_1 is an equivalence relation but not R_2

(2) R_2 is an equivalence relation but not R_1

(3) both R_1 and R_2 are equivalence relations

(4) neither R_1 nor R_2 is an equivalence relation **RT0103**

10. For $\alpha \in N$ consider a relation R on N given by $R = \{(x, y) : 3x + \alpha y$ is a multiple of $7\}$. The relation R is an equivalence relation if and only if : **[JEE(Main) 2022]**

(1) $\alpha = 14$

(2) α is a multiple of 4

(3) 4 is the remainder when α is divided by 10

(4) 4 is the remainder when α is divided by 7 **RT0104**

11. Let R be a relation from the set $\{1, 2, 3, \ldots, 60\}$ to itself such that $R = \{(a,b) : b = pq$, where $p, q \geq 3$ are prime numbers$\}$. Then, the number of elements in R is : **[JEE(Main) 2022]**

(1) 600 (2) 660 (3) 540 (4) 720 **RT0105**

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12. Let $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$. Then on N : **[JEE(Main) 2022]**

(1) Both R_1 and R_2 are equivalence relations

(2) Neither R_1 nor R_2 is an equivalence relation

(3) R_1 is an equivalence relation but R_2 is not

(4) R_2 is an equivalence relation but R_1 is not **RT0106**

13. Let R_1 and R_2 be relations on the set $\{1, 2, \ldots, 50\}$ such that $R_1 = \{(p, p^n) : p$ is a prime and $n \geq 0$ is an integer$\}$ and $R_2 = \{(p, p^n) : p$ is a prime and $n = 0$ or $1\}$. Then, the number of elements in $R_1 - R_2$ is _____. **[JEE(Main) 2022]**

RT0107

14. Let a set $A = A_1 \cup A_2 \cup \ldots \cup A_k$, where $A_i \cap A_j = \phi$ for $i \neq j$ $1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y) : y \in A_i$ if and only if $x \in A_i, 1 \leq i \leq k\}$. Then, R is :

[JEE(Main) 2022]

(1) reflexive, symmetric but not transitive

(2) reflexive, transitive but not symmetric

(3) reflexive but not symmetric and transitive

(4) an equivalence relation **RT0108**

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ANSWER KEY

Do yourself -1

1. R^{-1} : {(2,1), (4,3), (6,5)} **2.** D **3.** A

Do yourself -2

5. A **6.** [1] = {1, 2}, [2] = {2, 1}, [4] = {4, 5}

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A	A	C	C	A	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	D	D	C	B	B	D	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	C	B	B	D	C	D	C	A

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,C	A,B,D	A,C,D	A,B	A,B,C	A,B	A,B,C	A,D	B,C,D
Que.	11	12	13	14	15					
Ans.	A,B,D	A,B,C	B,C	A,C,D	A,B,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A	A	C	A,B,C	A,D	2.00	2.00	6.00	B	
Q.10	A	B	C	D						
	P,Q,R,S	P,Q	P,R	Q,R						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	64	4	9	64	2	20.00	5.00	4096.00	2.00	1.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	1	3	1	2	4	4	4	4
Que.	11	12	13	14						
Ans.	2	2	8	4						

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EXERCISE (O-1)

Straight Objective Type

Arithmetic mean, weighted mean, Combined mean

1. Mean of the first n terms of the A.P. a, (a + d), (a + 2d), ... is-

(A) $a+\frac{nd}{2}$ (B) $a+\frac{(n-1)d}{2}$ (C) a + (n – 1) d (D) a + nd

SI0001

2. The A.M. of ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ...${}^nC_n$ is -

(A) $\frac{2^n}{n}$ (B) $\frac{2^{n+1}}{n}$ (C) $\frac{2^n}{n+1}$ (D) $\frac{2^{n+1}}{n+1}$

SI0003

3. The mean of 9 terms is 15. if one new term is added and mean become 16, then the value of new term is :-

(A) 23 (B) 25 (C) 27 (D) 30

SI0006

4. If the mean of first n natural numbers is equal to $\frac{n+7}{3}$, then n is equal to-

(A) 10 (B) 11 (C) 12 (D) none of these

SI0007

5. The mean of first three terms is 14 and mean of next two terms is 18. The mean of all the five terms is-

(A) 15.5 (B) 15.0 (C) 15.2 (D) 15.6

SI0008

6. The mean of a set of numbers is $\bar{x}$. If each number is decreased by λ, the mean of the new set is-

(A) $\bar{x}$ (B) $\bar{x}+\lambda$ (C) $\lambda-\bar{x}$ (D) $\bar{x}-\lambda$

SI0010

7. The mean of 50 observations is 36. If its two observations 30 and 42 are deleted, then the mean of the remaining observations is-

(A) 48 (B) 36 (C) 38 (D) none of these

SI0011

8. The A.M. of n observation is $\bar{x}$. If the sum of n – 4 observations is K, then the mean of remaining observations is-

(A) $\frac{\bar{x}-K}{4}$ (B) $\frac{n\bar{x}-K}{n-4}$ (C) $\frac{n\bar{x}-K}{4}$ (D) $\frac{n\bar{x}-(n-4)K}{4}$

SI0013

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9. The sum of squares of deviation of variates from their A.M. is always :-

(A) Zero (B) Minimum

(C) Maximum (D) Nothing can be said

SI0015

10. The weighted mean of first n natural numbers when their weights are equal to corresponding natural number, is :-

(A) $\frac{n+1}{2}$ (B) $\frac{2n+1}{3}$ (C) $\frac{(n+1)(2n+1)}{6}$ (D) None of these

SI0018

11. In a group of students, the mean weight of boys is 65 kg. and mean weight of girls is 55 kg. If the mean weight of all students of group is 61 kg, then the ratio of the number of boys and girls in the group is :-

(A) 2 : 3 (B) 3 : 1 (C) 3 : 2 (D) 4 : 3

SI0020

Median, Mode

12. The median of the numbers 6, 14, 12, 8, 10, 9, 11, is :-

(A) 8 (B) 10 (C) 10.5 (D) 11

SI0022

13. Median of the following freq. dist.

x_i	3	6	10	12	7	15
f_i	3	4	2	8	13	10

(A) 7 (B) 10 (C) 8.5 (D) None of these

SI0023

14. Median is independent of change of :-

(A) only Origin (B) only Scale

(C) Origin and scale both (D) Neither origin nor scale

SI0024

15. A series which have numbers three 4's, four 5's, five 6's, eight 7's, seven 8's and six 9's then the mode of numbers is :-

(A) 9 (B) 8 (C) 7 (D) 6

SI0025

16. Mode of the following freqency distribution

x :	4	5	6	7	8
f :	6	7	10	8	3

(A) 5 (B) 6 (C) 8 (D) 10

SI0026

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17. The mode of the following freq. dist is :-

Class	1–10	11–20	21–30	31–40	41–50
f_i	5	7	8	6	4

(A) 24 (B) 23.83 (C) 27.16 (D) None of these

SI0027

Symmetric and asymmetric distribution, Range

18. For a normal dist :-

(A) mean = median (B) median = mode

(C) mean = mode (D) mean = median = mode

SI0028

19. The range of observations 2, 3, 5, 9, 8, 7, 6, 5, 7, 4, 3 is :-

(A) 6 (B) 7 (C) 5.5 (D) 11

SI0030

Mean Deviation

20. Mean deviation of the observations 70, 42, 63, 34, 44, 54, 55, 46, 38, 48 from median is :-

(A) 7.8 (B) 8.6 (C) 7.6 (D) 8.8

SI0033

21. Mean deviation of 5 observations from their mean 3 is 1.2, then coefficient of mean deviation is :-

(A) 0.24 (B) 0.4 (C) 2.5 (D) None of these

SI0034

22. The mean deviation from median is

(A) greater than the mean deviation from any other central value

(B) less than or equal to the mean deviation from any other central value

(C) equal to the mean deviation from any other central value

(D) maximum if all values are positive

SI0035

Variance and Standard Deviation

23. The S.D. of the first n natural numbers is-

(A) $\sqrt{\frac{n^2-1}{2}}$ (B) $\sqrt{\frac{n^2-1}{3}}$ (C) $\sqrt{\frac{n^2-1}{4}}$ (D) $\sqrt{\frac{n^2-1}{12}}$

SI0037

24. The variance of observations 112, 116, 120, 125, 132 is :-

(A) 58.8 (B) 48.8 (C) 61.8 (D) None of these

SI0038

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25. If $\sum_{i=1}^{10}(x_i-15)=12$ and $\sum_{i=1}^{10}(x_i-15)^2=18$ then the S.D. of observations x_1, x_2 ..... x_{10} is :-

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) None of these

SI0039

26. The variance of series a, a + d, a + 2d, ..., a + 2nd is :-

(A) $\frac{n(n+1)}{2}d^2$ (B) $\frac{n(n+1)}{3}d^2$ (C) $\frac{n(n+1)}{6}d^2$ (D) $\frac{n(n+1)}{12}d^2$

SI0041

27. Variance is independent of change of-

(A) only origin (B) only scale

(C) origin and scale both (D) none of these

SI0042

28. If each observation of a dist., whose variance is σ^2, is multiplied by λ, then the S.D. of the new new observations is-

(A) σ (B) $\lambda\sigma$ (C) $|\lambda|\sigma$ (D) $\lambda^2\sigma$

SI0046

Probability Distribution

29. In binomial probability distribution, mean is 3 and standard deviation is $\frac{3}{2}$. Then the probability distribution is

(A) $\left(\frac{3}{4}+\frac{1}{4}\right)^{12}$ (B) $\left(\frac{1}{4}+\frac{3}{4}\right)^{12}$ (C) $\left(\frac{1}{4}+\frac{3}{4}\right)^{9}$ (D) $\left(\frac{3}{4}+\frac{1}{4}\right)^{9}$ **SI0140**

30. If the mean and variance of a binomial variate X are 2 and 1 respectively, then the probability that X takes a value greater than 1, is

(A) $\frac{2}{3}$ (B) $\frac{4}{5}$ (C) $\frac{7}{8}$ (D) $\frac{15}{16}$ **SI0141**

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EXERCISE (O-2)

Multiple Correct Answer Type

1. If the mean of n observations $1^2, 2^2, 3^2, \ldots\ldots n^2$ is $\frac{46n}{11}$, then n is less than -

(A) 11 (B) 12 (C) 23 (D) 22

SI0049

2. The average age of a group of men and women is 30years. If average age of men is 32 and that of women is 27, then the percentage of women in the group is greater than or equal to -

(A) 60 (B) 50 (C) 40 (D) 30

SI0051

3. Mean and median of four numbers a, b, c and d ($b < a < d < c$) is 35 and 25 respectively then the value of $b + c - a - d$ will be divisible by :-

(A) 8 (B) 5 (C) 40 (D) 20

SI0052

4. The A.M. of first n terms of the series 1.3.5, 3.5.7, 5.7.9,....., is $an^3 + bn^2 + cn + d$ then -

(A) $a + b = 10$ (B) $a - b = 6$ (C) $a + b + c + d = 15$ (D) $a - d = 4$

SI0055

5. The observations 29, 32, 48, 50, x, x + 2, 72, 78, 84, 95 are arranged in ascending order and their median is 63 then the value of x is less than :-

(A) 61 (B) 62 (C) 62.5 (D) 63

SI0056

6. If the mode of a distribution is 18 and the mean is 24, then median is greater than -

(A) 18 (B) 24 (C) 22 (D) 21

SI0057

7. For 15 observations of x, mean and median were found to be 12 and 20 respectively. Later an observation which was 25 found to be wrong then replaced by its correct value 55, if new mean and median are a and b respectively, then :-

(A) $a + b = 65$ (B) $ab = 280$ (C) $a - b = -6$ (D) $b = a + 6$

SI0062

8. The mean square deviation about –1 and +1 of a set of observations are 7 and 3 respectively then standard deviation of the set is greater than or equal to :-

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) 1

SI0068

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9. The S.D. of the following freq. dist. is greater than or equal to :-

Class	0 – 10	10 – 20	20 – 30	30 – 40
f_i	1	3	4	2

(A) 7.8 (B) 9 (C) 8.1 (D) 0.9

SI0074

10. The S.D. of the numbers 31, 32, 33, ... 47 is less than or equal to -

(A) $2\sqrt{6}$ (B) $4\sqrt{3}$ (C) $\sqrt{\frac{47^2-1}{12}}$ (D) $2\sqrt{5}$

SI0082

11. The median and standard deviation (S.D.) of a distribution will be, If each term is increased by 2 -

(A) S.D. will increased by 2 (B) median will increased by 2

(C) median will remain same (D) S.D. will remain same

SI0084

12. The coefficient of mean deviation from median of observations 40, 62, 54, 90, 68, 76 is less than or equal to :-

(A) 2.16 (B) 0.2 (C) 5 (D) 0.1

SI0087

13. A group of 10 observations has mean 5 and S.D. $2\sqrt{6}$. another group of 20 observations has mean 5 and S.D. $3\sqrt{2}$, then the S.D. of combined group of 30 observations is less than or equal to :-

(A) $\sqrt{5}$ (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

SI0088

14. x_i 's are observations such that $(x_i^2 - 7x_i + 12)(x_i^3 - x_i^2 - 4x_i + 4) = 0$, then :-

(A) Mean = $\frac{8}{5}$ (B) Median = 2 (C) Variance = $\frac{106}{25}$ (D) Mean = 2

SI0091

15. Which of the following is/are correct ?

(A) $\sigma^2 = \frac{\sum_{i=1}^{n}(x_i-1)^2}{n} - \left(\frac{\sum_{i=1}^{n}(x_i-1)}{n}\right)^2$ (B) $\sigma = \frac{\sum_{i=1}^{n}x_i^2}{n} - \left(\frac{\sum_{i=1}^{n}x_i}{n}\right)^2$

(C) $\sigma^2 = \frac{\sum_{i=1}^{n}f_i(x_i-\bar{x})^2}{\sum_{i=1}^{n}f_i}$ (D) $\sigma^2 = \frac{\sum_{i=1}^{n}f_i(x_i-k)^2}{\sum_{i=1}^{n}f_i} - (\bar{x})^2$

SI0142

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16. A die is tossed thrice. If getting a four is considered a success, then the mean and variance of the probability distribution of the number of successes are **SI0143**

(A) Mean = $\frac{1}{2}$ (B) Variance = $\frac{5}{12}$ (C) Mean = $\frac{5}{6}$ (D) Variance = $\frac{1}{12}$

17. Which of the following can not be obtained from graph ?

(A) Mean (B) Median (C) Mode (D) Variance **SI0144**

18. If X is binomial variate with parameters n and p where $0 < p < 1$ such that $\frac{P(X = r)}{P(X = n - r)}$ is independent of n and r, then solution of $|x - 1| = p$

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{4}$ (D) $\frac{3}{4}$ **SI0145**

19. If X follows a binomial distribution with parameter $n = 8$ and $p = \frac{1}{2}$. If $P(|X - 4| \le 2) = m$ then

(A) $[m] = 0$ (where [.] is greatest integer function)

(B) Arithmetic mean of $(m - 1)$ and $\left(\frac{m}{7}\right)$ is $\frac{1}{32}$

(C) The number of solutions of the equation $\cos x = m$ in $\left[-2\pi, \frac{3\pi}{2}\right]$ is 3

(D) None of these **SI0146**

20. Let X denote the number of times heads occur in n tosses of a fair coin. If P(X = 4), P(X = 5) and P(X = 6) are in A.P., then the value of n can be

(A) 10 (B) 14 (C) 12 (D) 7 **SI0147**

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

If the variance of $x_1, x_2, ..., x_n$ is σ^2 then

var. of $x_1 \pm \lambda, x_2 \pm \lambda, ..., x_n \pm \lambda$ is σ^2

var. of $\lambda x_1, \lambda x_2, ..., \lambda x_n$ is $\lambda^2\sigma^2$

1. If each observation of a dist. whose S.D. is σ, is increased by λ, then the variance of the new observations is -

(A) σ (B) $\sigma + \lambda$ (C) σ^2 (D) $\sigma^2 + \lambda$

SI0148

2. The standard deviation of variate x_i is σ. Then standard deviation of the variate $\frac{ax_i + b}{c}$, where a , b, c are constants is-

(A) $\left(\frac{a}{c}\right)\sigma$ (B) $\left|\frac{a}{c}\right|\sigma$ (C) $\left(\frac{a^2}{c^2}\right)\sigma$ (D) None of these

SI0149

Paragraph for Question 3 to 5

If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then

3. Then $\alpha + \beta$ is divisible by

(A) 10 (B) 2 (C) 5 (D) 15 **SI0150**

4. Then equation $x^2 - (\alpha + \beta + 1)x + \alpha\beta + \alpha = 0$ has roots

(A) 5 (B) –4 (C) 6 (D) –5 **SI0151**

5. Mean of $\alpha\beta$, 8, 10, $\alpha^2 + \beta$, 2, 6 is less than

(A) 14 (B) 12 (C) 15 (D) 18 **SI0152**

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Paragraph for Question 6 and 7

A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarised it in the table given below

Number of cars	frequency
0 – 10	7
10 – 20	14
20 – 30	13
30 – 40	12
40 – 50	20
50 – 60	11
60 – 70	15
70 – 80	8

6. Find mean of the data **SI0153**

7. Find mode of the data

Answer Q.8 and Q.9 by appropriately matching the information given in the two List of the following table

	List-I		List-II
(I)	The weighted mean of first n natural numbers whose weights are equal, is	(P)	$\frac{n^2-1}{3}$
(II)	Variance of the group $\alpha, \alpha+2, \alpha+4, \alpha+6, \ldots\ldots$ upto n terms ($\alpha \neq 0$) is	(Q)	$\frac{n}{2}$
(III)	The mean of values 0, 1, 2, ..., n when their weights are $1, {}^nC_1, {}^nC_2, \ldots, {}^nC_n$, resp., is	(R)	$\frac{n-1}{2}$
(IV)	The mean of n values of a distribution is $\bar{x}$. If its first value is increased by 1, second by 2, ... then the mean of new values will increase by	(S)	$\frac{n+1}{3}$
		(T)	$\frac{n+1}{2}$

SI0154

8. Which of the following is the only CORRECT combination ?

(A) (I), (S) (B) (II), (Q) (C) (I), (T) (D) (IV), (Q) **SI0155**

9. Which of the following is the only CORRECT combination ?

(A) (I), (P) (B) (II), (T) (C) (III), (S) (D) (IV), (T) **SI0156**

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10. Consider the following distribution of profits earned by 30 shops.

Profit (Rs. in lakhs)	Number of shops
1 – 10	2
10 – 20	10
20 – 30	7
30 – 40	7
40 – 50	4

	Column-I		**Column-II**
(A)	Mean	(P)	17.27
(B)	Mode	(Q)	136.73
(C)	Median	(R)	25.33
(D)	Variance	(S)	24.28

SI0157

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EXERCISE (O-4)

Numerical Grid Type

1. If the mean of following feq. dist. is 2.6, then the value of f is :-

x_i	1	2	3	4	5
f_i	5	4	f	2	3

SI0016

2. If the coefficient of variation and standard deviation of a distribution are 50% and 20 respectively, then its mean is-

SI0043

3. The mean deviation of the numbers 1, 2, 3, 4, 5 is-

SI0069

4. The median of 21 observation is 40. if each observations greater than the median are increased by 6, then the median of the observations will be-

SI0072

5. The mean of a dist. is 4. if its coefficient of variation is 58%. Then the S.D. of the dist. is:-

SI0075

6. The average age of a teacher and three students is 20 years. If all students are of equal age and the difference between the age of the teacher and that of a student is 20 years, then the age (in years) of the teacher is-

SI0077

7. The mean deviation from mean of observations 5, 10, 15, 20, ...85 is :-

SI0080

8. If standard deviation of variate x_i is 10, then variance of the variate $(50 + 5x_i)$ will be-

SI0081

9. The median of 19 observations of a group is 30. If two observations with values 8 and 32 are further included, then the median of the new group of 21 observation will be

SI0086

10. Median of 5 observations i.e. $3^{\log_9 4}, 5^{\log_{1/2} 8}, e^{2\ell n 3}, \ell n\left(\frac{1}{e^2}\right)+3, e^{2\ell n 3+\frac{1}{\log_4 e}}$:-

SI0078

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EXERCISE (JM)

1. All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given ? **[JEE(Main)-2013]**

(1) mean (2) median (3) mode (4) variance

SI0092

2. The variance of first 50 even natural numbers is :- **[JEE(Main)-2014]**

(1) $\frac{833}{4}$ (2) 833 (3) 437 (4) $\frac{437}{4}$

SI0093

3. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3, 4 and 5 are added to the data, then the mean of the resultant data, is :

[JEE(Main)-2015]

(1) 15.8 (2) 14.0 (3) 16.8 (4) 16.0

SI0094

4. If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true ? **[JEE(Main)-2016]**

(1) $3a^2 - 23a + 44 = 0$ (2) $3a^2 - 26a + 55 = 0$

(3) $3a^2 - 32a + 84 = 0$ (4) $3a^2 - 34a + 91 = 0$

SI0095

5. If $\sum_{i=1}^{9}(x_i - 5) = 9$ and $\sum_{i=1}^{9}(x_i - 5)^2 = 45$, then the standard deviation of the 9 items $x_1, x_2, ..., x_9$ is -

[JEE(Main)-2018]

(1) 4 (2) 2 (3) 3 (4) 9

SI0096

6. 5 students of a class have an average height 150 cm and variance 18 cm^2. A new student, whose height is 156 cm, joined them. The variance (in cm^2) of the height of these six students is:

[JEE(Main)-19]

(1) 22 (2) 20 (3) 16 (4) 18

SI0097

7. A data consists of n observations : $x_1, x_2, ..., x_n$. If $\sum_{i=1}^{n}(x_i + 1)^2 = 9n$ and $\sum_{i=1}^{n}(x_i - 1)^2 = 5n$, then the standard deviation of this data is : **[JEE(Main)-19]**

(1) 5 (2) $\sqrt{5}$ (3) $\sqrt{7}$ (4) 2

SI0098

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8. The mean of five observations is 5 and their variance is 9.20. If three of the given five observations are 1, 3 and 8, then a ratio of other two observations is : **[JEE(Main)-19]**

(1) 4 : 9 (2) 6 : 7 (3) 5 : 8 (4) 10 : 3

SI0099

9. The outcome of each of 30 items was observed; 10 items gave an outcome $\frac{1}{2}$ – d each, 10 items gave outcome $\frac{1}{2}$ each and the remaining 10 items gave outcome $\frac{1}{2}$ + d each. If the variance of this outcome data is $\frac{4}{3}$ then |d| equals :- **[JEE(Main)-19]**

(1) 2 (2) $\frac{\sqrt{5}}{2}$ (3) $\frac{2}{3}$ (4) $\sqrt{2}$

SI0100

10. The mean and the variance of five observation are 4 and 5.20, respectively. If three of the observations are 3, 4 and 4; then then absolute value of the difference of the other two observations, is : **[JEE(Main)-19]**

(1) 1 (2) 3 (3) 7 (4) 5

SI0101

11. The mean and variance of seven observations are 8 and 16, respectively. If 5 of the observations are 2, 4, 10, 12, 14, then the product of the remaining two observations is : **[JEE(Main)-19]**

(1) 40 (2) 49 (3) 48 (4) 45

SI0102

12. If the standard deviation of the numbers –1, 0, 1, k is $\sqrt{5}$ where $k > 0$, then k is equal to

(1) $2\sqrt{\frac{10}{3}}$ (2) $2\sqrt{6}$ (3) $4\sqrt{\frac{5}{3}}$ (4) $\sqrt{6}$

SI0103

13. If for some $x \in R$, the frequency distribution of the marks obtained by 20 students in a test is : **[JEE(Main)-19]**

Marks	2	3	5	7
Frequencey	$(x+1)^2$	$2x-5$	x^2-3x	x

then the mean of the marks is :

(1) 2.8 (2) 3.2 (3) 3.0 (4) 2.5

SI0104

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14. If both the mean and the standard deviation of 50 observations $x_1, x_2....,x_{50}$ are equal to 16, then the mean of $(x_1 - 4)^2, (x_2 - 4)^2,...(x_{50} - 4)^2$ is : **[JEE(Main)-19]**

(1) 525 (2) 380 (3) 480 (4) 400

SI0105

15. If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then x·y is equal to_____

[JEE(Main)-20]

SI0106

16. If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then m + n is equal to _____. **[JEE(Main)-20]**

SI0107

17. The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11. Then the correct variance is **[JEE(Main)-20]**

(1) 3.99 (2) 3.98 (3) 4.02 (4) 4.01

SI0108

18. The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 resepectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to **[JEE(Main)-20]**

(1) –20 (2) 10 (3) –10 (4) –5

I0109

19. Let the observations $x_i (1 \leq i \leq 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_i - 5) = 10$ and $\sum_{i=1}^{10}(x_i - 5)^2 = 40$. If μ and λ are the mean and the variance of the observations, $x_1 - 3, x_2 - 3, ..., x_{10} - 3$, then the ordered pair (μ, λ) is equal to : **[JEE(Main)-20]**

(1) (6, 6) (2) (3, 6) (3) (6, 3) (4) (3, 3)

SI0110

20. Consider three observations a, b and c such that b = a + c. If the standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true ? **[JEE(Main)-21]**

(1) $b^2 = 3(a^2 + c^2) + 9d^2$ (2) $b^2 = a^2 + c^2 + 3d^2$

(3) $b^2 = 3(a^2 + c^2 + d^2)$ (4) $b^2 = 3(a^2 + c^2) - 9d^2$ **SI0158**

21. Let in a series of 2n observations, half of them are equal to a and remaining half are equal to –a. Also by adding a constant b in each of these observations, the mean and standard deviation of new set become 5 and 20, respectively. Then the value of $a^2 + b^2$ is equal to : **[JEE(Main)-21]**

(1) 425 (2) 650 (3) 250 (4) 925 **SI0159**

22. If the variance of 10 natural numbers 1, 1, 1,...., 1, k is less than 10, then the maximum possible value of k is _____. **[JEE(Main)-21]**

SI0160

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23. Let $X_1, X_2, ..., X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is _____. **[JEE(Main)-21]**

SI0161

24. Consider the statistics of two sets of observations as follows :

	Size	Mean	Variance
Observation I	10	2	2
Observation II	n	3	1

If the variance of the combined set of these two observations is $\frac{17}{9}$, then the value of n is equal to _____. **[JEE(Main)-21]**

SI0162

25. Consider a set of 3n numbers having variance 4. In this set, the mean of first 2n numbers is 6 and the mean of the remaining n numbers is 3. A new set is constructed by adding 1 into each of first 2n numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k, then 9k is equal to _____. **[JEE(Main)-21]**

SI0163

26. The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is_. **[JEE(Main)-21]**

SI0164

27. The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to : **[JEE(Main)-22]**

(A) 10 (B) 36 (C) 43 (D) 60 **SI0165**

28. The mean of the numbers a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to : **[JEE(Main)-22]**

(A) 60 (B) 55 (C) 50 (D) 45 **SI0166**

29. The number of values of $a \in \mathbb{N}$ such that the variance of 3, 7, 12 a, 43 – a is a natural number is: **[JEE(Main)-22]**

(A) 0 (B) 2 (C) 5 (D) infinite **SI0167**

30. Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively. If the mean and variance of the first 4 observation are $\frac{7}{2}$ and a respectively, then $(4a + x_5)$ is equal to: **[JEE(Main)-22]**

(A) 13 (B) 15 (C) 17 (D) 18 **SI0168**

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EXERCISE (JA)

1. Consider the given data with frequency distribution

x_i	3	8	11	10	5	4
f_i	5	2	3	2	4	4

Match each entry in **List-I** to the correct entries in **List-II**.

List-I		**List-II**	
(P)	The mean of the above data is	(1)	2.5
(Q)	The median of the above data is	(2)	5
(R)	The mean deviation about the mean of the above data is	(3)	6
(S)	The mean deviation about the median of the above data is	(4)	2.7
		(5)	2.4

The correct option is : **[JEE(Advanced)-2023]**

(A) (P) → (3) (Q) → (2) (R) → (4) (S) → (5)

(B) (P) → (3) (Q) → (2) (R) → (1) (S) → (5)

(C) (P) → (2) (Q) → (3) (R) → (4) (S) → (1)

(D) (P) → (3) (Q) → (3) (R) → (5) (S) → (5)

SI0169

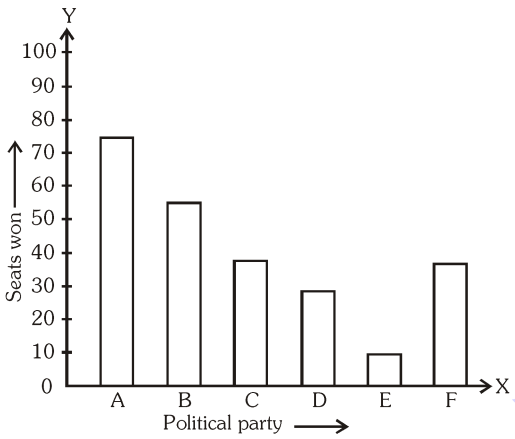
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ANSWER KEY

Do yourself - 1

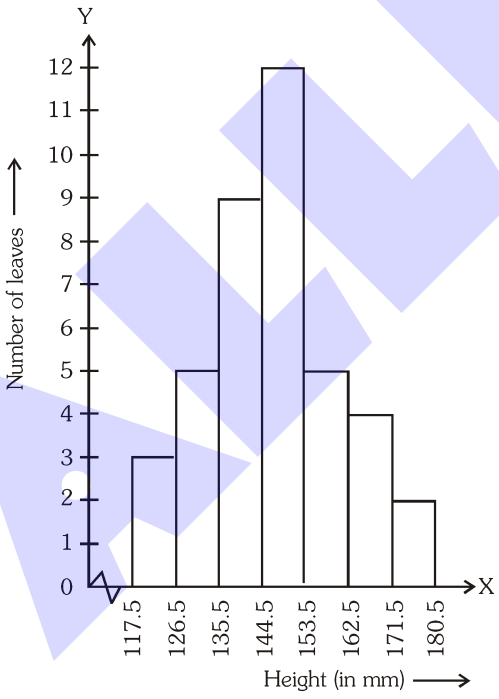
1. (i) The required bar graph is given below :

Scale : on y-axis 1 unit = 10 seats.



(ii) The political party A won the maximum number of seats.

2. (i) The required histogram of the above frequency distribution is as shown here :



(ii) Yes, other suitable graphical representation is a "frequency polygon"

(iii) No, it is not a correct statement. The maximum number of leaves are not 153 mm long rather they are from 145 mm to 153 mm long.

Do yourself - 2

1. B **2.** B **3.** A **4.** B **5** A

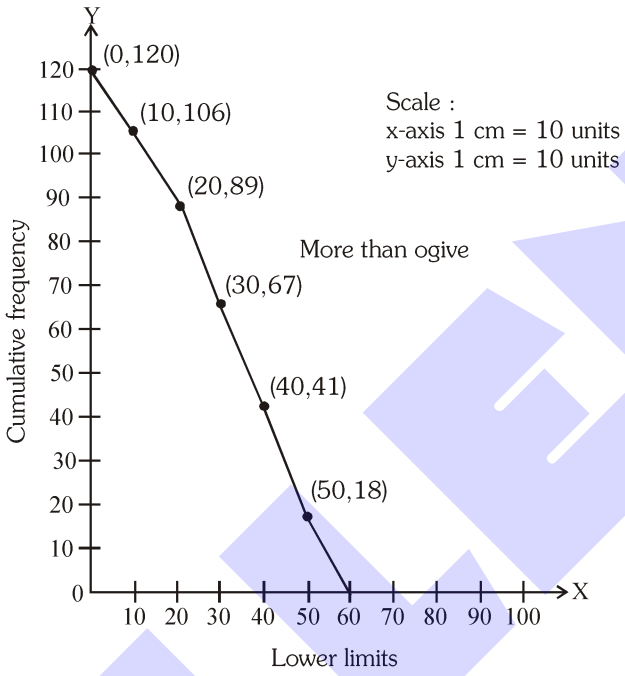
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Do yourself - 3

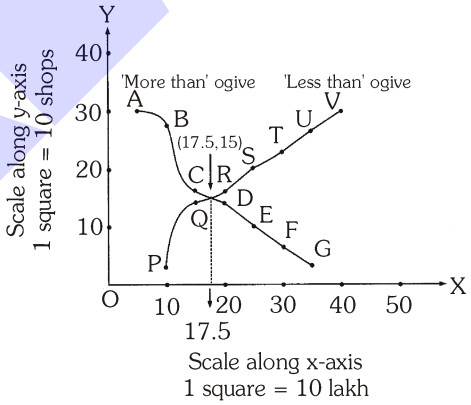
1.

Weight (in kg)	Cumulative Frequency
More than or equal to 0	120
More than or equal to 10	106
More than or equal to 20	89
More than or equal to 30	67
More than or equal to 40	41
More than or equal to 50	18

Plotting the points :



2. Join these points by a freehand to get ogives for 'more than' type and 'less than' type.



The two ogives intersect each other at point (17.5, 15).

Hence, the median profit is Rs. 17.5 lakhs.

Do yourself - 4

1. The class 165.5-168.5 has maximum frequency, so it is the modal class.

2. f = 10.

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Do yourself - 5

1. C **2.** C **3.** C **4.** B

5. A **6.** A

Do yourself - 6

1. C.V. in weaights is greater than the C.V. in heights

Therefore, we can say that weights show more variability than heights.

2. B **3.** Y

Do yourself - 7

1. B **2.** A **3.** A **4.** B **5.** A

6. C **7.** C **8.** B **9.** B

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	B	D	D	B	C	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	C	D	C	B	B	D	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	D	B	B	B	A	C	A	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C,D	C,D	A,B,C,D	A,C,D	C,D	A,D	B,C,D	A,B,D	A,B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	A,B,C	B,C,D	A,B,C	A,C	A,B	A,D	A,B	A,B,C	B,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	B	A,B,C	A,C	A,C,D	40.70	44.70	C	D	
Q.10	A	B	C	D						
	R	P	S	Q						

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EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	40	1.20	40	2.32	35	21.17	2500	30.00	9.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	2	3	2	2	2	1	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	1	4	54	18	1	1	4	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	11	4	5	2	35	C	A	A	B

EXERCISE # JEE-ADVANCED

Que.	1	
Ans.	A	

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CHECK YOUR GRASP MATHEMATICAL REASONING EXERCISE-I

1. The inverse of the statement $(p \wedge \sim q) \rightarrow r$ is-
(1) $\sim(p \vee \sim q) \rightarrow \sim r$ (2) $(\sim p \wedge q) \rightarrow \sim r$
(3) $(\sim p \vee q) \rightarrow \sim r$ (4) None of these
MR0001

2. $(\sim p \vee \sim q)$ is logically equivalent to-
(1) $p \wedge q$ (2) $\sim p \rightarrow q$
(3) $p \rightarrow \sim q$ (4) $\sim p \rightarrow \sim q$
MR0002

3. The equivalent statement of $(p \leftrightarrow q)$ is-
(1) $(p \wedge q) \vee (p \vee q)$
(2) $(p \rightarrow q) \vee (q \rightarrow p)$
(3) $(\sim p \vee q) \vee (p \vee \sim q)$
(4) $(\sim p \vee q) \wedge (p \vee \sim q)$
MR0003

4. If the compound statement $p \rightarrow (\sim p \vee q)$ is false then the truth value of p and q are respectively-
(1) T, T (2) T, F (3) F, T (4) F, F
MR0004

5. The statement $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ is-
(1) a tautology
(2) a contradiction
(3) neither a tautology nor a contradiction
(4) None of these
MR0005

6. Negation of the statement $(p \wedge r) \rightarrow (r \vee q)$ is-
(1) $\sim(p \wedge r) \rightarrow \sim(r \vee q)$
(2) $(\sim p \vee \sim r) \vee (r \vee q)$
(3) $(p \wedge r) \wedge (r \wedge q)$
(4) $(p \wedge r) \wedge (\sim r \wedge \sim q)$
MR0006

7. The dual of the statement $\sim p \wedge [\sim q \wedge (p \vee q) \wedge \sim r]$ is-
(1) $\sim p \vee [\sim q \vee (p \vee q) \vee \sim r]$
(2) $p \vee [q \vee (\sim p \wedge \sim q) \vee r]$
(3) $\sim p \vee [\sim q \vee (p \wedge q) \vee \sim r]$
(4) $\sim p \vee [\sim q \wedge (p \wedge q) \wedge \sim r]$
MR0007

8. Which of the following is correct-
(1) $(\sim p \vee \sim q) \equiv (p \wedge q)$
(2) $(p \rightarrow q) \equiv (\sim q \rightarrow \sim p)$
(3) $\sim(p \rightarrow \sim q) \equiv (p \wedge \sim q)$
(4) $\sim(p \leftrightarrow q) \equiv (p \rightarrow q) \vee (q \rightarrow p)$
MR0008

9. The contrapositive of $p \rightarrow (\sim q \rightarrow \sim r)$ is-
(1) $(\sim q \wedge r) \rightarrow \sim p$ (2) $(q \rightarrow r) \rightarrow \sim p$
(3) $(q \vee \sim r) \rightarrow \sim p$ (4) None of these
MR0009

10. The converse of $p \rightarrow (q \rightarrow r)$ is-
(1) $(q \wedge \sim r) \vee p$ (2) $(\sim q \vee r) \vee p$
(3) $(q \wedge \sim r) \wedge \sim p$ (4) $(q \wedge \sim r) \wedge p$
MR0010

11. If p and q are two statement then $(p \leftrightarrow \sim q)$ is true when-
(1) p and q both are true
(2) p and q both are false
(3) p is false and q is true
(4) None of these
MR0011

12. Statement $(p \wedge q) \rightarrow p$ is-
(1) a tautology (2) a contradiction
(3) neither (1) nor (2) (4) None of these
MR0012

13 If statements p, q, r have truth values T, F, T respectively then which of the following statement is true-
(1) $(p \rightarrow q) \wedge r$ (2) $(p \rightarrow q) \vee \sim r$
(3) $(p \wedge q) \vee (q \wedge r)$ (4) $(p \rightarrow q) \rightarrow r$
MR0013

14. If statement $p \rightarrow (q \vee r)$ is true then the truth values of statements p, q, r respectively-
(1) T, F, T (2) F, T, F
(3) F, F, F (4) All of these
MR0014

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15. Which of the following statement is a contradiction-
(1) $(p \wedge q) \wedge (\sim(p \vee q))$ (2) $p \vee (\sim p \wedge q)$
(3) $(p \to q) \to p$ (4) $\sim p \vee \sim q$

MR0015

16. The negative of the statement "If a number is divisible by 15 then it is divisible by 5 or 3"
(1) If a number is divisible by 15 then it is not divisible by 5 and 3
(2) A number is divisible by 15 and it is not divisible by 5 or 3
(3) A number is divisible by 15 or it is not divisible by 5 and 3
(4) A number is divisible by 15 and it is not divisible by 5 and 3

MR0016

17. If $x = 5$ and $y = -2$ then $x - 2y = 9$. The contrapositive of this statement is-
(1) If $x - 2y \neq 9$ then $x \neq 5$ or $y \neq -2$
(2) If $x - 2y \neq 9$ then $x \neq 5$ and $y \neq -2$
(3) If $x - 2y = 9$ then $x = 5$ and $y = -2$
(4) None of these

MR0017

18. The negation of the statement "$2 + 3 = 5$ and $8 < 10$" is-
(1) $2 + 3 \neq 5$ and $8 \not< 10$
(2) $2 + 3 \neq 5$ or $8 > 10$
(3) $2 + 3 \neq 5$ or $8 \geq 10$
(4) None of these

MR0018

19. For any three simple statement p, q, r the statement $(p \wedge q) \vee (q \wedge r)$ is true when-
(1) p and r true and q is false
(2) p and r false and q is true
(3) p, q, r all are false
(4) q and r true and p is false

MR0019

20. Which of the following statement is a tautology-
(1) $(\sim p \vee \sim q) \vee (p \vee \sim q)$
(2) $(\sim p \vee \sim q) \wedge (p \vee \sim q)$
(3) $\sim p \wedge (\sim p \vee \sim q)$
(4) $\sim q \wedge (\sim p \vee \sim q)$

MR0020

21. Which of the following statement is a contradiction-
(1) $(\sim p \vee \sim q) \vee (p \vee \sim q)$
(2) $(p \to q) \vee (p \wedge \sim q)$
(3) $(\sim p \wedge q) \wedge (\sim q)$
(4) $(\sim p \wedge q) \vee (\sim q)$

MR0021

22. The negation of the statement $q \vee (p \wedge \sim r)$ is equivalent to-
(1) $\sim q \wedge (p \to r)$ (2) $\sim q \wedge \sim(p \to r)$
(3) $\sim q \wedge (\sim p \wedge r)$ (4) None of these

MR0022

23. Let Q be a non empty subset of N. and q is a statement as given below :-
q : There exists an even number $a \in Q$.
Negation of the statement q will be :-
(1) There is no even number in the set Q.
(2) Every $a \in Q$ is an odd number.
(3) (1) and (2) both
(4) None of these

MR0023

24. The statement $\sim(p \to q) \leftrightarrow (\sim p \vee \sim q)$ is-
(1) a tautology
(2) a contradiction
(3) neither a tautology nor a contradiction
(4) None of these

MR0024

25. Which of the following is equivalent to $(p \wedge q)$
(1) $p \to \sim q$ (2) $\sim(\sim p \wedge \sim q)$
(3) $\sim(p \to \sim q)$ (4) None of these

MR0025

26. The dual of the following statement "Reena is healthy and Meena is beautiful" is-
(1) Reena is beaufiful and Meena is healthy
(2) Reena is beautiful or Meena is healthy
(3) Reena is healthy or Meena is beutiful
(4) None of these

MR0026

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27. If p is any statement, t and c are a tautology and a contradiction respectively then which of the following is not correct-

(1) $p \wedge t \equiv p$ (2) $p \wedge c \equiv c$

(3) $p \vee t \equiv c$ (4) $p \vee c \equiv p$

MR0027

28. If S*(p, q) is the dual of the compound statement S(p, q) then S*(~p, ~q) is equivalent to-

(1) S(~p, ~q) (2) ~S(p, q)

(3) ~S*(p, q) (4) None of these

MR0028

29. If p is any statement, t is a tautology and c is a contradiction then which fo the following is not correct-

(1) $p \wedge (\sim c) \equiv p$

(2) $p \vee (\sim t) \equiv p$

(3) $t \vee c \equiv p \vee t$

(4) $(p \wedge t) \vee (p \vee c) \equiv (t \wedge c)$

MR0029

30. If p, q, r are simple statement with truth values T, F, T respectively then the truth value of $((\sim p \vee q) \wedge \sim r) \rightarrow p$ is-

(1) True

(2) False

(3) True if r is false

(4) True if q is true

MR0030

31. Which of the following is wrong-

(1) $p \vee \sim p$ is a tautology

(2) $\sim(\sim p) \leftrightarrow p$ is a tautology

(3) $p \wedge \sim p$ is a contradiction

(4) $((p \wedge p) \rightarrow q) \rightarrow p$ is a tautology

MR0031

32. The statement "If $2^2 = 5$ then I get first class" is logically equivalent to-

(1) $2^2 = 5$ and I do not get first class

(2) $2^2 = 5$ or I do not get first class

(3) $2^2 \neq 5$ or I get first class

(4) None of these

MR0032

33. If statement $(p \vee \sim r) \rightarrow (q \wedge r)$ is false and statement q is true then statement p is-

(1) true (2) false

(3) may be true or false (4) None of these

MR0033

34. Which of the following statement are not logically equivalent-

(1) $\sim(p \vee \sim q)$ and $(\sim p \wedge q)$

(2) $\sim(p \rightarrow q)$ and $(p \wedge \sim q)$

(3) $(p \rightarrow q)$ and $(\sim q \rightarrow \sim p)$

(4) $(p \rightarrow q)$ and $(\sim p \wedge q)$

MR0034

35. Consider the following statements

p : Virat kohli plays cricket.

q : Virat kohli is good at maths

r : Virat kohli is successful.

then negation of the statement "If virat kohli plays cricket and is not good at maths then he is successful" will be :-

(1) $\sim p \wedge (q \wedge r)$ (2) $(\sim p \vee q) \wedge r$

(3) $p \wedge (\sim q \wedge \sim r)$ (4) None of these

MR0035

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36. Let p statement "If 2 > 5 then earth will not rotate" and q be the statement "$2 \not> 5$ or earth will not rotate".

Statement–1 : p and q are equivalent.

Statement–2 : $m \to n$ and $\sim m \vee n$ are equivalent.

(1) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1.

(2) Statement–1 is false, Statement–2 is true.

(3) Statement–1 is true, Statement–2 is false.

(4) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1.

MR0036

37. Which of the following is a tautology :-

(1) $[(\sim p \wedge p) \to q] \longrightarrow (p \wedge p)$

(2) $[(\sim p \wedge p) \to q] \longrightarrow (\sim p \to p)$

(3) $[(\sim p \wedge p) \to q] \longrightarrow (p \to p)$

(4) None of these

MR0037

38. Negation of the statement "No one in the class is fond of music" is :-

(1) everyone in the class is fond of music.

(2) Some of the students in the class are fond of music.

(3) There exists a student in the class who is fond of music.

(4) (2) and (3) both

MR0038

CHECK YOUR GRASP **ANSWER-KEY** **EXERCISE-I**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	2	2	4	3	2	1	1	3	1	4	4	1	4	1	3	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38		
Ans.	3	1	3	3	3	3	3	2	4	1	4	3	3	4	3	4	3	4		

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PREVIOUS YEAR QUESTIONS MATHEMATICAL REASONING EXERCISE-II

1. The statement $\sim(p \leftrightarrow \sim q)$ is : **[JEE(Main)-2014]**

(1) equivalent to $p \leftrightarrow q$

(2) equivalent to $\sim p \leftrightarrow q$

(3) a tautology

(4) a fallacy

MR0041

2. The negation of $\sim s \vee (\sim r \wedge s)$ is equivalent to : **[JEE(Main)-2015]**

(1) $s \vee (r \vee \sim s)$ (2) $s \wedge r$

(3) $s \wedge \sim r$ (4) $s \wedge (r \wedge \sim s)$

MR0042

3. The Boolean Expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to :- **[JEE(Main)-2016]**

(1) $p \vee \sim q$ (2) $\sim p \wedge q$ (3) $p \wedge q$ (4) $p \vee q$

MR0043

4. The following statement $(p \rightarrow q) \rightarrow [(\sim p \rightarrow q) \rightarrow q]$ is : **[JEE(Main)-2017]**

(1) a fallacy

(2) a tautology

(3) equivalent to $\sim p \rightarrow q$

(4) equivalent to $p \rightarrow \sim q$

MR0044

5. The Boolean expression $\sim (p \vee q) \vee (\sim p \wedge q)$ is equivalent to : **[JEE(Main)-2018]**

(1) p (2) q (3) ~q (4) ~p

MR0045

6. If the Boolean expression $(p \oplus q) \wedge (\sim p \odot q)$ is equivalent to $p \wedge q$, where $\oplus, \odot \in \{\wedge, \vee\}$, then the ordered pair $(\oplus, \odot)$ is: **[JEE(Main)-19]**

(1) $(\wedge, \vee)$ (2) $(\vee, \vee)$ (3) $(\wedge, \wedge)$ (4) $(\vee, \wedge)$

MR0046

7. The logical statement

$$[\sim(\sim p \vee q) \vee (p \wedge r)] \wedge (\sim q \wedge r)$$

is equivalent to : **[JEE(Main)-19]**

(1) $(p \wedge r) \wedge \sim q$ (2) $(\sim p \wedge \sim q) \wedge r$

(3) $\sim p \vee r$ (4) $(p \wedge \sim q) \vee r$

MR0047

8. Consider the following three statements :

P : 5 is a prime number.

Q : 7 is a factor of 192.

R : L.C.M. of 5 and 7 is 35.

Then the truth value of which one of the following statements is true ? **[JEE(Main)-19]**

(1) $(P \wedge Q) \vee (\sim R)$ (2) $(\sim P) \wedge (\sim Q \wedge R)$

(3) $(\sim P) \vee (Q \wedge R)$ (4) $P \vee (\sim Q \wedge R)$

MR0048

9. If q is false and $p \wedge q \leftrightarrow r$ is true, then which one of the following statements is a tautology? **[JEE(Main)-19]**

(1) $(p \vee r) \rightarrow (p \wedge r)$ (2) $p \vee r$

(3) $p \wedge r$ (4) $(p \wedge r) \rightarrow (p \vee r)$

MR0049

10. Contrapositive of the statement "If two numbers are not equal, then their squares are not equal." is :- **[JEE(Main)-19]**

(1) If the squares of two numbers are equal, then the numbers are equal.

(2) If the squares of two numbers are equal, then the numbers are not equal.

(3) If the squares of two numbers are not equal, then the numbers are equal.

(4) If the squares of two numbers are not equal, then the numbers are not equal.

MR0050

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11. The contrapositive of the statement "If you are born in India, then you are a citizen of India", is : **[JEE(Main)-19]**

(1) If you are born in India, then you are not a citizen of India.

(2) If you are not a citizen of India, then you are not born in India.

(3) If you are a citizen of India, then you are born in India.

(4) If you are not born in India, then you are not a citizen of India.

MR0051

12. For any two statements p and q, the negation of the expression $p\vee(\sim p\wedge q)$ is **[JEE(Main)-19]**

(1) $p\wedge q$ (2) $p\leftrightarrow q$

(3) $\sim p\vee\sim q$ (4) $\sim p\wedge\sim q$

MR0052

13. If the truth value of the statement $p\rightarrow(\sim q\vee r)$ is false(F), then the truth values of the statements p, q, r are respectively : **[JEE(Main)-19]**

(1) F, T, T (2) T, F, F

(3) T, T, F (4) T, F, T

MR0053

14. The Boolean expression $\sim(p\Rightarrow(\sim q))$ is equivalent to : **[JEE(Main)-19]**

(1) $(\sim p)\Rightarrow q$ (2) $p\vee q$

(3) $q\Rightarrow\sim p$ (4) p ^ q

MR0054

15. Let A, B, C and D be four non-empty sets. The contrapositive statement of "If $A\subseteq B$ and $B\subseteq D$, then $A\subseteq C$" is : **[JEE(Main)-20]**

(1) If $A\subseteq C$, then $B\subset A$ or $D\subset B$

(2) If $A\not\subseteq C$, then $A\not\subseteq B$ or $B\not\subseteq D$

(3) If $A\not\subseteq C$, then $A\subseteq B$ and $B\subseteq D$

(4) If $A\not\subseteq C$, then $A\not\subseteq B$ and $B\subseteq D$

MR0055

16. The logical statement $(p\Rightarrow q)\wedge(q\Rightarrow\sim p)$ is equivalent to : **[JEE(Main)-20]**

(1) p (2) q (3) ~p (4) ~q

MR0056

17. Which of the following statements is a tautology? **[JEE(Main)-20]**

(1) $\sim(p\vee\sim q)\rightarrow p\vee q$

(2) $\sim(p\wedge\sim q)\rightarrow p\vee q$

(3) $\sim(p\vee\sim q)\rightarrow p\wedge q$

(4) $p\vee(\sim q)\rightarrow p\wedge q$

MR0057

18. Which one of the following is a tautology ? **[JEE(Main)-20]**

(1) $P\wedge(P\vee Q)$

(2) $P\vee(P\wedge Q)$

(3) $Q\rightarrow(P\wedge(P\rightarrow Q))$

(4) $(P\wedge(P\rightarrow Q))\rightarrow Q$

MR0058

19. If $p\rightarrow(p\wedge\sim q)$ is false, then the truth values of p and q are respectively : **[JEE(Main)-20]**

(1) F, T (2) T, T (3) F, F (4) T, F

MR0059

20. Negation of the statement : **[JEE(Main)-20]**

$\sqrt{5}$ is an integer or 5 is irrational is :

(1) $\sqrt{5}$ is irrational or 5 is an integer.

(2) $\sqrt{5}$ is not an integer and 5 is not irrational.

(3) $\sqrt{5}$ is an integer and 5 is irrational.

(4) $\sqrt{5}$ is not an integer or 5 is not irrational.

MR0060

21. Consider the two statements :

(S1) : $(p\rightarrow q)\vee(\sim q\rightarrow p)$ is a tautology.

(S2) : $(p\wedge\sim q)\wedge(\sim p\vee q)$ is a fallacy.

Then : **[JEE(Main)-21]**

(1) only (S1) is true.

(2) both (S1) and (S2) are false.

(3) both (S1) and (S2) are true.

(4) only (S2) is true. **MR0066**

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22. The contrapositive of the statement "If you will work, you will earn money" is :

(1) You will earn money, if you will not work

(2) If you will earn money, you will work

(3) If you will not earn money, you will not work

(4) To earn money, you need to work

[JEE(Main)-21]

MR0067

23. Which of the following statement is a tautology? **[JEE(Main)-22]**

(A) $((\sim q)\wedge p)\wedge q$

(B) $((\sim q)\wedge p)\wedge(p\wedge(\sim p))$

(C) $((\sim q)\wedge p)\vee(p\vee(\sim p))$

(D) $(p\wedge q)\wedge(\sim(p\wedge q))$

MR0068

24. Let p, q, r be three logical statements. Consider the compound statements

$S_1:((\sim p)\vee q)\vee((\sim p)\vee r)$ and

$S_2:p\rightarrow(q\vee r)$

Then, which of the following is NOT true ?

(A) If S_2 is True, then S_1 is True

(B) If S_2 is False, then S_1 is False

(C) If S_2 is False, then S_1 is True

(D) If S_1 is False, then S_2 is False

[JEE(Main)-22]

MR0069

25. The statement $(p\Rightarrow q)\vee(p\Rightarrow r)$ is NOT equivalent to:

(A) $(p\wedge(\sim r))\Rightarrow q$

(B) $(\sim q)\Rightarrow((\sim r)\vee p)$

(C) $p\Rightarrow(q\vee r)$

(D) $(p\wedge(\sim q))\Rightarrow r$

[JEE(Main)-22]

MR0070

PREVIOUS YEARS QUESTIONS ANSWER-KEY EXERCISE-II

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	4	2	4	1	1	4	4	1	2	4	3	4	2
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	3	1	4	2	2	3	3	C	C	B					

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